

# PhD Math Camp

## Calculus II: Integrals

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# Introduction

- In this lecture we will learn what **integrals** are and how are they useful.
- But we will motivate integrals from a specific angle: **probability**.
  - We will learn that integrals help compute probabilities for continuous variables.
  - Also, for continuous variables, integrals define important properties (aka, *parameters*). For example, the true mean of a population is defined with an integral.
- Integrals show up in many applications, but fortunately computers will do the calculations. *We just need to know what an integral is, and what does it mean when it shows up in a specific application.*

# Sample Space, Events, Random Variables, and Probability

- Sample space,  $\Omega$ , includes all possible outcomes of a random process.
- An event is a subset of the sample space, e.g., all elements  $x_i \in \Omega$  where a property  $Q(x_i)$  is true.
- A random variable (RV),  $X$ , is numerical and represents a collection of outcomes of a random process (an event).  $X$  can be discrete (integer subset) or continuous (real subset).
- The probability of an event is the count of favorable outcomes (i.e., the event happens) divided by all outcomes.
- Generally, for any discrete RV,  $X$ , then

$$P(X = x_i) = \frac{|\{\omega_j \in \Omega : \omega_j \leftrightarrow X = x_i\}|}{|\Omega|} = \frac{\text{\#Favorable Outcomes}}{\text{\#All Outcomes}}$$

# Discrete vs. Continuous Random Variables

- Discrete Random Variable (DRV):
  - Let  $X$  be a DRV, then  $X$  takes on values from a subset of integers,  $\mathbb{Z}$ .
  - Probability function is defined as,  $f : X \rightarrow [0, 1]$ , such that for each  $x_i \in X$ , the probability function  $f(x_i)$  gives the probability that  $X$  equals  $x_i$ .
- Continuous Random Variable (CRV):
  - Let  $X$  be a CRV, then  $X$  takes on infinitely many values from a real subset.
  - Probability function is defined as,  $f : C \subseteq X \rightarrow [0, 1]$ , such that for any subset  $C = [a, b] = \{x_i \in X : x_i \in [a, b]\}$ , the probability function  $f(C)$  yields the probability that  $X$  lies in  $C$ .

# From the Probability Distribution Function to the Cumulative Distribution Function (CDF)

## The CDF:

- The Cumulative Distribution Function (CDF) calculates the probability of obs.  $x_i$  or a lower value. It sums all probabilities up to  $x_i$ . For a Discrete RV:

$$\Pr(X \leq x_i) = CDF(x_i) = F(x_i) = \sum_{k=-\infty}^i f(x_k)$$

- Thus, starting from a Probability Distribution Function, we can always construct the Cumulative Distribution Function by adding up the probabilities from the minimum possible value of  $X$  up to the probability of  $x_i$

# From the Cumulative Distribution Function (CDF) to the Probability Distribution Function

## Deriving Probability from CDF:

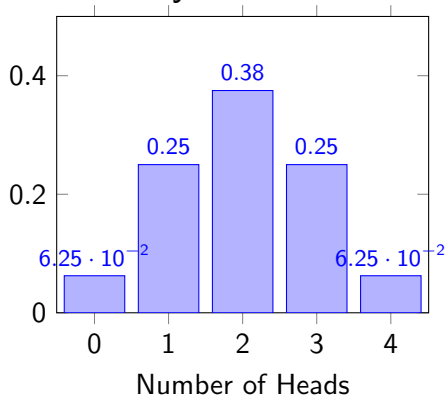
- With the CDF of a random variable, you can find its Probability Distribution Function.
- For a Discrete RV, simply subtract the CDF values at two successive points to find the probability of the latter point:

$$F(x_i) - F(x_{i-1}) = \sum_{k=-\infty}^i f(x_k) - \sum_{k=-\infty}^{i-1} f(x_k) = f(x_i)$$

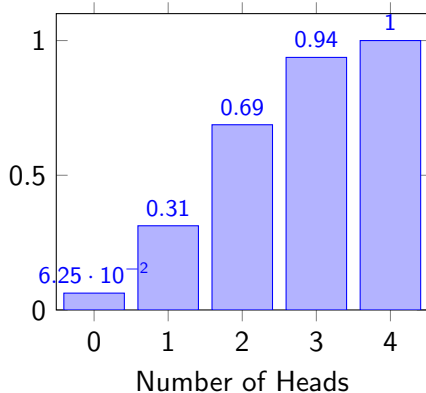
- This allows you to calculate probability at a point from accumulated probability.

# Probability Distribution and Cumulative Distribution of the Total Number of Heads When Throwing Four Coins

## Probability Distribution



## Cumulative Distribution



## Key Takeaway: Interplay of the Probability Distribution and the CDF

- When you look at a discrete Probability Distribution Function and CDF, the Probability Distribution at  $x = a + 1$  equals exactly the rate of change in accumulated probability when one moves from  $a$  to  $a + 1$  in the CDF.
- Metaphorically, the Probability Distribution seems to represent the “speed” at which we accumulate probability.
- Similarly, the CDF value at a given point  $a$ , is the sum of all probability “speeds” from the minimum possible value of  $X$  to  $a$ .
- This is a nice connection: **summing the areas under the Probability Distribution graph yields the CDF, and the “slope” of the CDF graph recovers the Probability Distribution.**



# Continuous Random Variables and the Probability Density Function (PDF)

- *But*, continuous random variables have an important difference – **the probabilities are no longer points, but areas under a curve.**
- An infinitely thin slice in continuous space contributes no area, so to compute probabilities we use closed intervals,  $[a, b]$ , and a function,  $f(x_i)$  that computes the *relative likelihood* of  $x_i \in [a, b] \subseteq X$ .
- We call  $f$ , the Probability **Density** Function (PDF). If  $f$  is the PDF of a Continuous RV  $X$ , then  $f$  needs to satisfy two conditions:
  1. The PDF is always non-negative, i.e.  $f(x) \geq 0$  for all  $x$  in the domain of  $X$ .
  2. The area under the entire PDF curve is 1.

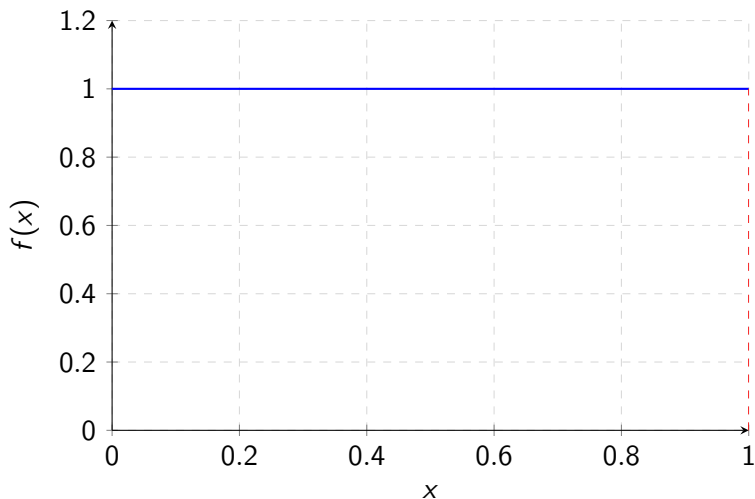
# Uniform Distribution for Continuous Random Variables

- Consider the uniform distribution on the interval  $[0, 1]$ .
- The PDF is:

$$f(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- The total area under the PDF curve is 1.

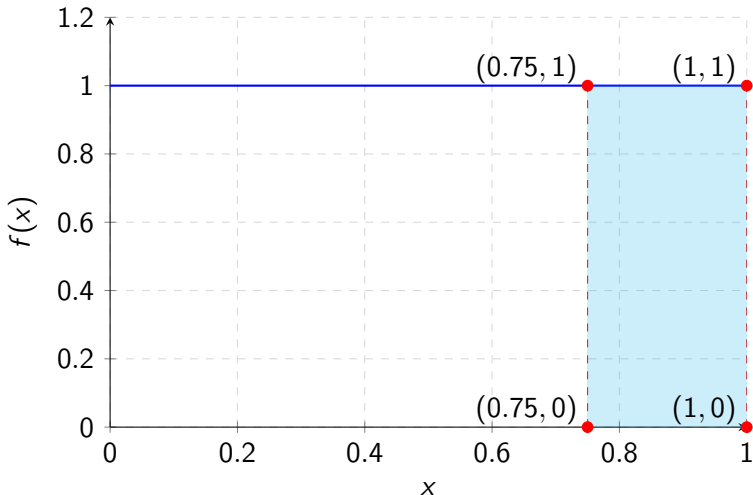
## Graph of the Uniform Distribution PDF



# Probability Calculation for Uniform Distribution

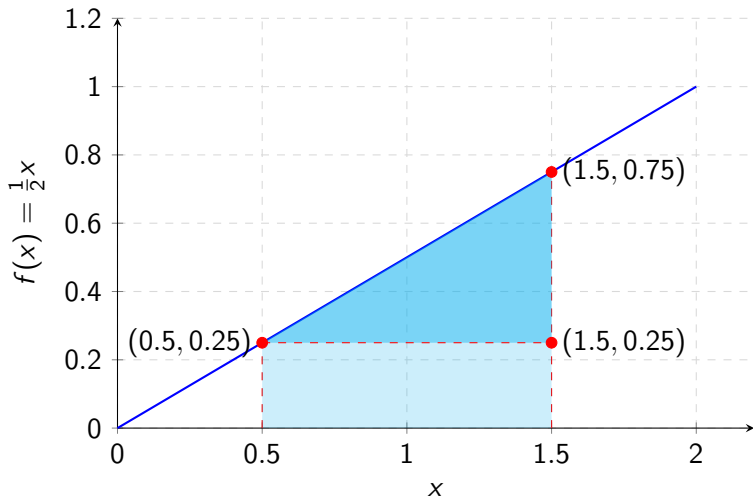
- The probability that  $X$  is greater than 0.75 in a uniform distribution from 0 to 1 is calculated as follows:

$$P(0.75 < X) = 1 \times (1 - 0.75) = 0.25$$



## Another Example: a Triangular PDF

- Suppose  $f(x_i) = (1/2)x_i \leftrightarrow x_i \in [0, 2]$  and  $f(x_i) = 0 \leftrightarrow x_i \notin [0, 2]$ .
- To calculate the probability that  $X$  lies between 0.5 and 1.5 we can use one rectangle and one triangle to represent the area.



## How do we compute areas under a curve?

- So, given the previous definition, for any Continuous RV  $X$  with PDF  $f(x_i)$ , to calculate the probability that  $X$  lies in some interval  $[a, b]$  we need to compute the area under the PDF curve between  $a$  and  $b$ .
- How do we do that for *any* curve  $f(x_i)$ ?

# Approximating Area: Riemann Sums

- **Riemann sums** are a method to approximate the area under a curve by breaking it down into simpler geometric shapes, such as rectangles.
- To compute a Riemann sum, we:
  1. Divide the interval  $[a, b]$  into  $n$  subintervals of equal width,  $\Delta x = \frac{b-a}{n}$ .
  2. On each subinterval, construct a rectangle. The height of each rectangle is determined by the function value at a specific point within the subinterval.
  3. Sum the areas of all rectangles to get an approximation of the total area under the curve.

# Approximating Area: Riemann Sums

- The Riemann sum can be written as:

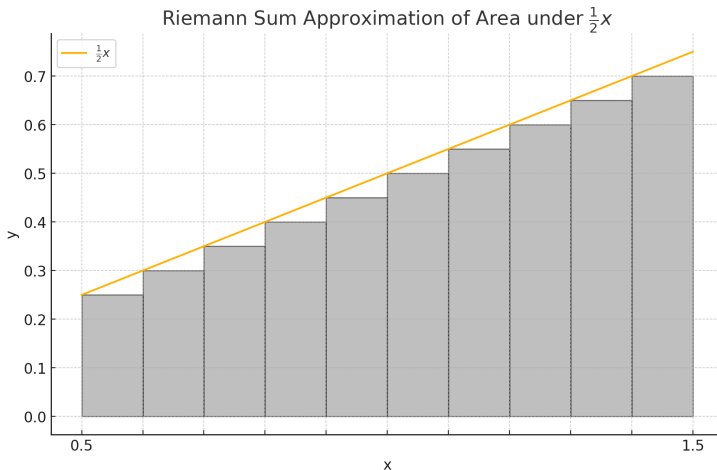
$$S_n = \sum_{i=1}^n f(x_i^*) \cdot \frac{b-a}{n} = \sum_{i=1}^n f(x_i^*) \Delta x$$

where  $x_i^*$  is a chosen point in the  $i$ -th subinterval.

- As  $n$  increases (i.e., more rectangles), the approximation becomes more accurate, and in the limit as  $n \rightarrow \infty$ , the Riemann sum converges to the exact area, which is the integral of the function.

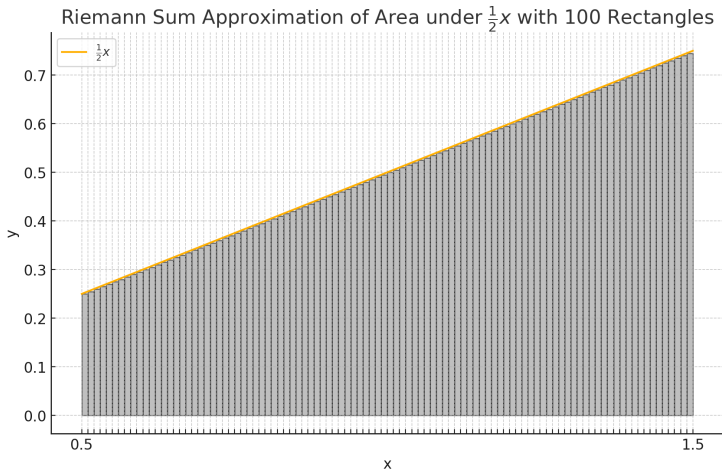


# Triangular PDF: Riemann Approximation with 10 Rectangles



- Riemann Sum = 0.475. Exact Value = 0.50.

# Triangular PDF: Riemann Approximation with 100 Rectangles



- Riemann Sum = 0.4975. Exact Value = 0.50.

# The Riemann Integral: Taking the Limit

- While Riemann sums provide an approximation, the **Riemann integral** gives the exact area under the curve.
- The Riemann integral is defined as the limit of the Riemann sum as the number of subintervals  $n$  approaches infinity:

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x = \lim_{\|\Delta x\| \rightarrow 0} \sum_{i=1}^n f(x_i^*) \Delta x_i$$

- Here:
  - $\Delta x = \frac{b-a}{n}$  is the width of each subinterval.
  - $x_i^*$  is a chosen point in the  $i$ -th subinterval  $[x_{i-1}, x_i]$ .
  - The sum  $\sum_{i=1}^n f(x_i^*) \Delta x$  represents the total area of the rectangles approximating the area under  $f(x)$ .
- The integral  $\int_a^b f(x) dx$  represents the exact area

## Key Takeaway: Riemann Integrals and Their Challenges

- **Riemann integrals** provide a clear and intuitive way to understand the area under a curve by summing up the areas of rectangles and taking the limit as the number of rectangles approaches infinity.
- *Intuitive interpretation*: The integral represents the exact area under the curve  $f(x)$  between two points  $a$  and  $b$ .
- **However**, calculating the limit of these sums to find the exact area can be tedious and complex, especially for complicated functions.
- This complexity raises the question: Is there a more efficient way to compute these areas without directly taking limits?

# Introduction to Antiderivatives

- An **antiderivative** of a function  $f(x)$  is another function  $F(x)$  whose derivative is  $f(x)$ .
- In other words,  $F(x)$  satisfies:

$$F'(x) = f(x)$$

- *Example:* If  $f(x) = 2x$ , then one possible antiderivative is  $F(x) = x^2$  because:

$$\frac{d}{dx} (x^2) = 2x$$

- **Key Idea:** The process of finding an antiderivative is often called *integration*, and it is the reverse of differentiation.

# Antiderivatives and the Constant $C$

- When we find the antiderivative (indefinite integral) of a function  $f(x)$ , we obtain a **family of functions** rather than a single function.
- The general form of an antiderivative is given by:

$$F(x) + C$$

where  $C$  is an arbitrary constant.

- **Key Insight:** The constant  $C$  accounts for the fact that differentiation of a constant yields zero, meaning any function of the form  $F(x) + C$  has the same derivative  $f(x)$ .
- *Example:* For  $f(x) = 2x$ , the antiderivative is:

$$\int 2x \, dx = x^2 + C$$

where  $C$  could be any real number.

# Common Rules for Finding Indefinite Integrals

| Rule                           | Indefinite Integral                                                                                                                 |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Power Rule</b>              | $\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$                                                                           |
| <b>Constant Multiple Rule</b>  | $\int k \cdot f(x) dx = k \cdot \int f(x) dx$                                                                                       |
| <b>Sum/Difference Rule</b>     | $\int (f(x) \pm g(x)) dx = \int f(x) dx \pm \int g(x) dx$                                                                           |
| <b>Exponential Rule</b>        | $\int e^{kx} dx = \frac{1}{k} e^{kx} + C$                                                                                           |
| <b>Trigonometric Integrals</b> | $\begin{aligned}\int \sin(x) dx &= -\cos(x) + C \\ \int \cos(x) dx &= \sin(x) + C \\ \int \sec^2(x) dx &= \tan(x) + C\end{aligned}$ |
| <b>Logarithmic Rule</b>        | $\int \frac{1}{x} dx = \ln  x  + C \quad (x \neq 0)$                                                                                |

# The Fundamental Theorem of Calculus: Part 1

- The Fundamental Theorem of Calculus (FTC) connects differentiation with integration, offering a practical way to compute integrals.
- **FTC Part 1:** If  $F(x)$  is an antiderivative of  $f(x)$  on an interval  $[a, b]$ , then:

$$\frac{d}{dx}(F(x)) = \frac{d}{dx} \left( \int_a^x f(t) dt \right) = f(x)$$

- This means that the derivative of the integral function of  $f(x)$  (from  $a$  to  $x$ ) is equal to the original function  $f(x)$ .
- **Key Insight:** Differentiation and integration are inverse processes.



# The Fundamental Theorem of Calculus: Part 2

- **FTC Part 2:** If  $F(x)$  is any antiderivative of  $f(x)$ , then the definite integral of  $f(x)$  from  $a$  to  $b$  is:

$$\int_a^b f(x) dx = F(b) - F(a)$$

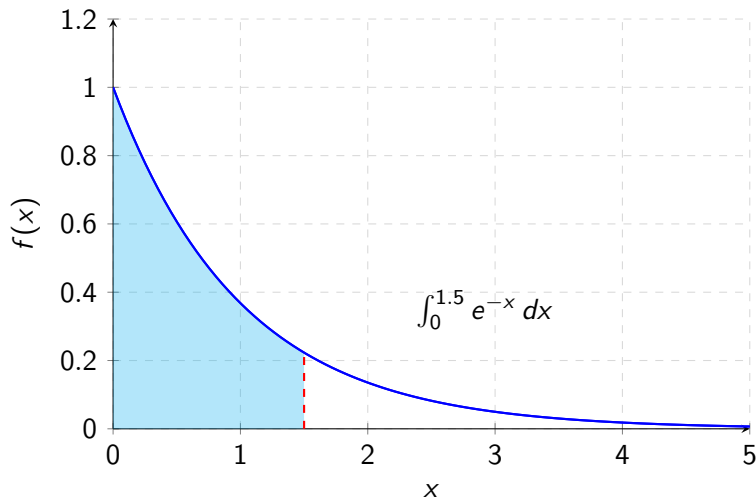
- **Key Insight:** Instead of using Riemann sums to find the area under  $f(x)$ , we can find an antiderivative  $F(x)$  and evaluate it at the endpoints  $a$  and  $b$ .
- *Example:* If  $f(x) = 2x$ , then:

$$\int_0^2 2x dx = [x^2]_0^2 = F(2) - F(0) = 2^2 - 0^2 = 4$$

## Example: Exponential Distribution

- Consider the exponential distribution with rate parameter  $\lambda = 1$ :

$$f(x) = e^{-x} \quad \text{for } x \geq 0$$



## Example: Computing Probabilities Using the Exponential Distribution

- Consider a random variable  $X$  that follows an **exponential distribution** with rate parameter  $\lambda > 0$ . The PDF is:

$$f(x) = \lambda e^{-\lambda x} \quad \text{for } x \geq 0$$

- Question:** What is the probability that  $X$  is less than or equal to a value  $a$ ?

$$P(X \leq a) = CDF(X) = \int_0^a \lambda e^{-\lambda x} dx$$

- Solution:**

$$\int_0^a \lambda e^{-\lambda x} dx = \left[ -e^{-\lambda x} \right]_0^a = 1 - e^{-\lambda a}$$

- Thus,  $P(X \leq a) = 1 - e^{-\lambda a}$ .

# The Normal Distribution

- **Normal Distribution:**

- Continuous probability distribution defined by mean  $\mu$  and standard deviation  $\sigma$ .
- PDF:

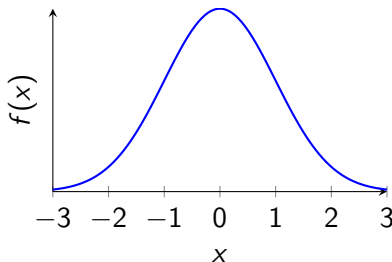
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- **Computing Probabilities:**

- Probability of an interval  $[a, b]$ :

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

- CDF:  $F(x) = \int_{-\infty}^x f(t) dt$



# Standard Normal Distribution

- **Standard Normal Distribution:**

- A special case of the normal distribution with mean  $\mu = 0$  and standard deviation  $\sigma = 1$ .
- The probability density function (PDF) is given by:

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

- **Computing Probabilities using Integrals:**

- For  $a \leq Z \leq b$ , the probability is:

$$P(a \leq Z \leq b) = \int_a^b \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

- The CDF denoted by  $\Phi(x)$ , is expressed as:

$$\Phi(x) = P(Z \leq x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$$

# Computing Probabilities for the Standard Normal Distribution

The area under the PDF from  $-\infty$  up to  $x$  defines the  $CDF(x) = P(Z \leq x)$ .

- [Interactive Demo](#)

# Derivatives, Integrals, and Probability

- **Calculating Changes Over Intervals:**

- To find the change in any  $F(x)$  from  $x = a$  to  $x = b$ , just compute  $F(b) - F(a)$ .
- If  $F(x)$  is the CDF, then
$$F(b) - F(a) = \text{CDF}(b) - \text{CDF}(a) = P(a < x < b).$$

- **PDF as a Derivative:**

- What if you only have the PDF  $f(x)$ ? Here's a twist: think of the PDF as the derivative of the CDF. Yep, that's right—PDF is the rate of change of probability for a tiny change in  $x$ .

- **Summing It All Up: The Integral:**

- To find  $P(a < x < b)$ , sum up all those tiny changes in probability (that is, integrate the PDF):

$$P(a < x < b) = \int_a^b f(x) dx = F(b) - F(a).$$

- Voilà! Integrals save the day by summing infinite tiny contributions!

# Expectation as the True Mean: The Center of Mass

- **True Mean (Expectation) of  $X$ :**

- The expectation  $\mathbb{E}[X]$  represents the true mean of a random variable  $X$ .
- It is defined as:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

where  $f(x)$  is the probability density function (PDF) of  $X$ .

- **Intuition: Center of Mass**

- Think of the expectation as the “center of mass” of the probability distribution.
- Just as the center of mass is the point where the mass of a physical object is balanced,  $\mathbb{E}[X]$  is the point where the distribution of probabilities balances out.
- This center balances all the possible values of  $X$ , weighted by their relative likelihoods (given by  $f(x)$ ).



# Numerical Integration: The Basics

- **Purpose:**

- Approximate the integral  $\int_a^b f(x) dx$  when an analytical solution is difficult or impossible.

- **Key Concept:**

- Break the interval  $[a, b]$  into  $n$  small subintervals of equal width  $\Delta x = \frac{b-a}{n}$ .
- Approximate the area under the curve  $f(x)$  on each subinterval.

- **Basic Approach:**

- The integral is approximated by summing up these small areas:

$$\int_a^b f(x) dx \approx \sum_{i=1}^n f(x_i) \cdot \Delta x$$

- $x_i$  represents the chosen point in the  $i$ -th subinterval.

## Example: Trapezoidal Rule for $\int_0^1 e^{-x^2} dx$

- **Goal:** Approximate  $\int_0^1 e^{-x^2} dx$  using the Trapezoidal Rule with  $n = 4$  subintervals.
- **Step 1: Subinterval Width:**  $\Delta x = \frac{1-0}{4} = 0.25$ .
- **Step 2: Function Values:**  $f(0) = 1$ ,  $f(0.25) \approx 0.939$ ,  $f(0.5) \approx 0.778$ ,  $f(0.75) \approx 0.569$ ,  $f(1) \approx 0.368$
- **Step 3: Apply Trapezoidal Rule:**

$$\int_0^1 e^{-x^2} dx \approx \frac{0.25}{2} [1 + 2(0.939 + 0.778 + 0.569) + 0.368]$$

$$\int_0^1 e^{-x^2} dx \approx 0.743$$

# Monte Carlo Integration: Step by Step

- **Step 1: Define the Integral.** Let  $\int_a^b f(x) dx$  be the integral you want to approximate.
- **Step 2: Generate Random Samples.** Generate  $N$  random points  $x_1, x_2, \dots, x_N$  uniformly in the interval  $[a, b]$ .
- **Step 3: Evaluate the Function.** Compute  $f(x_i)$  for each random sample  $x_i$ .
- **Step 4: Compute the Average.** Calculate the average value of  $f(x)$  over all samples:

$$\text{Average} = \frac{1}{N} \sum_{i=1}^N f(x_i)$$

- **Step 5: Estimate the Integral.** Multiply the average by the interval length  $(b - a)$  to estimate the integral:

$$\int_a^b f(x) dx \approx (b - a) \cdot \text{Average}$$

# Why Calculate the Average?

- **Intuition Behind the Average:**

- To estimate the area under  $f(x)$  over  $[a, b]$ , think of the average height of  $f(x)$  across this interval.
- The average of function values at sampled points gives a representative height:

$$\text{Average} = \frac{1}{N} \sum_{i=1}^N f(x_i)$$

- This average height, when multiplied by the interval length  $(b - a)$ , provides an approximation of the area:

$$\text{Approximate Area} = (b - a) \times \text{Average}$$

- If the samples are well-distributed, the average closely matches the true mean height of the function, leading to a good approximation of the integral.